

(1001CJA621120019)

Test Pattern



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2020 - 2021)

JEE(Advanced)
REVIEW TEST
08-11-2020

JEE(Main + Advanced) : NURTURE COURSE (PHASE : TEPS)

ANSWER KEY PAPER-1

PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A,C	A,B,C,D	A,B,D	B,C	A,B	A,B	D	C	A	A
SECTION-II	Q.	1	2	3	4	5	6	7	8		
	A.	476.20 to 476.60	3.00	6.92 to 6.93	6.80 to 6.90	1.41	1024.00	2.00	20.00		

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A,C,D	A,B,C,D	C,D	A,B,D	B	A,B,C	A	A	A	D
SECTION-II	Q.	1	2	3	4	5	6	7	8		
	A.	2.00	5.00	8.00	7.00	4.00	5.00	3.00	1.00		

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A,B,D	A,B,C	A,C,D	B,C,D	B	A,C	B	B	D	C
SECTION-II	Q.	1	2	3	4	5	6	7	8		
	A.	6.00	300.00	5.00	8.00	7.00	3.70	1.41 or 1.42	1.00		

(1001CJA621120020)

Test Pattern



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2020 - 2021)

JEE(Advanced)
REVIEW TEST
08-11-2020

JEE(Main + Advanced) : NURTURE COURSE (PHASE : TEPS)

ANSWER KEY PAPER-2

PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A,C,D	A,C,D	B,C,D	A,B	A,C	A,C	A	B	B	A
SECTION-II	Q.	1	2	3	4	5	6	7	8		
	A.	7.60	31.20	7.46	0.02 to 0.03	2.13	18.70	2.20	0.50		

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A,C	A,C	A,B,C,D	A	A,D	B,C	D	A	A	C
SECTION-II	Q.	1	2	3	4	5	6	7	8		
	A.	15.00	5.00	10.00	17.50	5.00	3.00	6.00	2.00		

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A,B,C	A,D	A,B,D	A,B,C	A,C,D	A,B,C,D	A	B	B	B
SECTION-II	Q.	1	2	3	4	5	6	7	8		
	A.	0.00	2.00	125.00	2.00	1.00	8.00	-1.00	24.00		



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2020 - 2021)

JEE(Advanced)

REVIEW TEST

08-11-2020

JEE(Main + Advanced) : NURTURE COURSE (PHASE : TEPS)

PAPER-1

PART-2 : CHEMISTRY

SOLUTION

SECTION-I

1. Ans. (A,C,D)
2. Ans. (A,B,C,D)
3. Ans. (C,D)
4. Ans. (A,B,D)

$$P_T = P_A^0 X_A + P_B^0 X_B$$

$$= P_A^0 + (P_B^0 - P_A^0) X_B$$

$$\Rightarrow 100 + 160 X_B$$

5. Ans. (B)
 $P_{\text{total}} = 0.92 \times 800 + 0.08 \times 300$
 $= 760 \text{ or } 1 \text{ atm} > P_{\text{obs.}}$
6. Ans. (A,B,C)
7. Ans. (A)
8. Ans. (A)
9. Ans. (A)
10. Ans. (D)

SECTION-II

1. Ans. (2.00)
2. Ans. (5.00)
3. Ans. (8.00)
4. Ans. (7.00)
5. Ans. (4.00)
6. Ans. (5.00)
7. Ans. (3.00)
8. Ans. (1.00)

$$\text{M.e. of } \text{MnO}_4^- = \frac{1}{2} \times 75 \times 10 = 375$$

$$\text{M.e. of } \text{Fe}^{2+} = 1 \times 150 \times 1 = 150$$

$$\text{M.e. of } \text{FeCrO}_4 = 1 \times 50 \times 1 = 50$$

PART-3 : MATHEMATICS

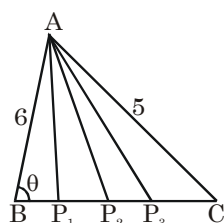
SOLUTION

SECTION-I

1. Ans. (A,B,D)

$$\text{Sol. } \cos \theta = \cos B = \frac{6^2 + 4^2 - 5^2}{2 \cdot 6 \cdot 4} = \frac{9}{16}$$

$$\cos C = \frac{4^2 + 5^2 - 6^2}{2 \cdot 4 \cdot 5} = \frac{1}{8}$$



$$AP_1^2 = 6^2 + 1^2 - 2 \cdot 1 \cdot 6 \cos \theta$$

$$AP_2^2 = 6^2 + 2^2 - 2 \cdot 2 \cdot 6 \cos \theta$$

$$AP_3^2 = 6^2 + 3^2 - 2 \cdot 3 \cdot 6 \cos \theta$$

$$AP_1^2 + AP_2^2 + AP_3^2$$

$$= 3 \cdot 6^2 + (1^2 + 2^2 + 3^2) - 2 \cdot 6 \cdot 6 \cos \theta = \frac{163}{2}$$

2. Ans. (A,B,C)

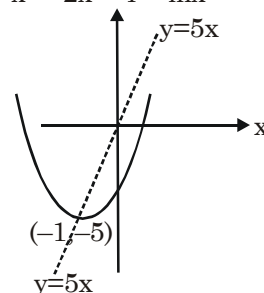
$$\text{Sol. } x^3 + ax^2 + bx + c$$

If roots are in A.P then $2a^3 - 9ab + 27c = 0$

3. Ans. (A,C,D)

$$x^2 + 2x - 4 = mx$$

Sol.



4. Ans. (B,C,D)

$$\sum_{\alpha=1}^{\infty} \frac{2^\alpha}{5^\alpha} = \frac{2}{5} + \left(\frac{2}{5}\right)^2 + \dots = \frac{2}{3}$$

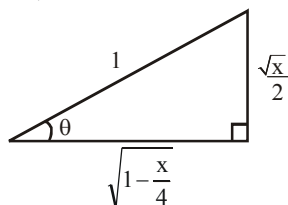
5. Ans. (B)

$$p = 4! \times {}^5C_3 \times {}^5C_3 \times \frac{3!}{2!} = 720$$

$$q = \frac{3!}{2!} \times 4! = 72$$

6. Ans. (A,C)

$$\sin^{-1} \left(\sqrt{1 - \frac{x}{4}} \right) = \theta \Rightarrow \sin \theta = \sqrt{1 - \frac{x}{4}}$$



$$\frac{\pi}{2} + \tan^{-1} y = \frac{2\pi}{3} \Rightarrow y = \frac{1}{\sqrt{3}}$$

$$\Rightarrow 0 \leq x \leq 4$$

$$\text{Max. value of } x^2 + y^2 \text{ is } \frac{49}{3}$$

$$\text{Min. value of } x^2 + y^2 \text{ is } \frac{1}{3}$$

Paragraph for Question 7 & 8

$$f(x) = 2x^2 - 6x + 5$$

for $f(x)$ to be one to one either

$$x \in \left[-100, \frac{3}{2}\right] \text{ or } x \in \left[\frac{3}{2}, 100\right]$$

but for longest subset set A has to be

$$\left[-100, \frac{3}{2}\right]$$

& for onto function set B = range =

$$\left[\frac{1}{2}, 20605\right]$$

7. Ans. (B)

For $g(x)$

$$y = 2x^2 - 6x + 5$$

$$2x^2 - 6x + (5 - y) = 0$$

$$x = \frac{6 \pm \sqrt{36 - 40 + 8y}}{4} = \frac{3 \pm \sqrt{2y - 1}}{2}$$

$$\text{but } x \leq \frac{3}{2},$$

$$\text{hence } g(x) = \frac{3 - \sqrt{2x - 1}}{2} = \frac{3}{2} - \sqrt{\frac{x}{2} - \frac{1}{4}}$$

& we can see if x increases $g(x)$ decreases.

8. Ans. (B)

For intersection points of $f(x)$ & $g(x)$

$$\text{we solve } f(x) = x \Rightarrow 2x^2 - 6x + 5 = x$$

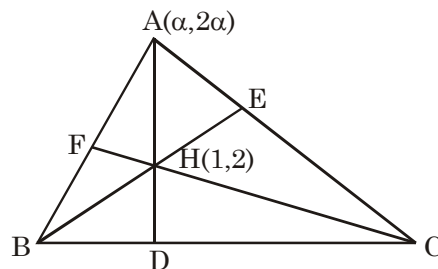
$$\Rightarrow x = 1 \text{ or } \frac{5}{2}$$

only $x = 1$ is in domain of $f(x)$ hence only one solution.

Paragraph for Question 9 and 10

9. Ans. (D)

Sol.



Equation of AC is $x - y + \alpha = 0$

equation of AB is $4x + y - 6\alpha = 0$

$$C\left(\frac{7-4\alpha}{3}, \frac{7-2\alpha}{3}\right)$$

$$B(2\alpha - 1, 4 - 2\alpha)$$

Let (h, k) be centroid

$$3h = \frac{7-4\alpha}{3} + 2\alpha - 1 + \alpha$$

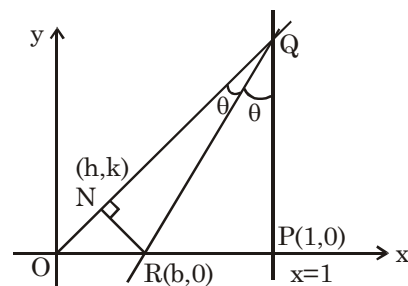
$$3k = \frac{7-2\alpha}{3} + 4 - 2\alpha + 2\alpha$$

$$\text{eliminating } \alpha \text{ we get } 9h = 4 + 5\alpha$$

$$9k = 19 - \alpha$$

$$(9x + 45y = 99)$$

10. Ans. (C)



Sol. Equation OQ is $kx - hy = 0$ (1)

$$RN = \frac{bk}{\sqrt{h^2 + k^2}}, RP = 1 - b$$

But $RN = RP$

$$\frac{bk}{\sqrt{h^2 + k^2}} = 1 - b$$

$$\frac{k}{\sqrt{h^2 + k^2}} = \frac{1}{b} - 1$$

$$\frac{k^2}{h^2 + k^2} = \frac{1}{b^2} - \frac{2}{b} + 1 \quad \text{.....(2)}$$

(slope of ON)(slope of RN) = 1

$$\Rightarrow \left(\frac{k}{h}\right)\left(\frac{-k}{b-h}\right) = -1 \quad \frac{k^2}{h(b-h)} = 1$$

$$b = \frac{h^2 + k^2}{h} \quad \text{.....(3)}$$

Put b is (2)

$$\frac{k^2}{h^2 + k^2} = \frac{h^2}{(h^2 + k^2)^2} - \frac{2h}{h^2 + k^2} + 1$$

$$\frac{h^2}{(h^2 + k^2)^2} - \frac{2h}{(h^2 + k^2)} + \frac{h^2}{(h^2 + k^2)} = 0$$

$$h^2 - 2h(h^2 + k^2) + h^2(h^2 + k^2) = 0$$

$$h(h + (h - 2)(h^2 + k^2)) = 0$$

$$x(x + (x - 2)(x^2 + y^2)) = 0$$

SECTION-II

1. **Ans. 6.00**

Sol. $(3^a \times 2^b) = (144)^c = 2^{4c} \cdot 3^{2c}$

$$\Rightarrow a = 2c; b = 4c$$

$$\frac{a+b}{c} = 6$$

2. **Ans. 300.00**

$$x + y + z = 47$$

$\Rightarrow$ The problem is similar to put 47 identical things in 3 diff. boxes, Let $x' = x - 1$

$$\Rightarrow x' + y + z = 2(23)$$

$$\Rightarrow {}^{23+3-1}C_{3-1} = {}^{25}C_2 = 300$$

3. **Ans. 5.00**

Sol. $\alpha + \beta = 3, \alpha\beta = 5$

$$(\alpha + \beta)^5 = (\alpha^5 + \beta^5) + 5\alpha\beta(\alpha^3 + \beta^3) + 10\alpha^2\beta^2(\alpha + \beta)$$

$$\Rightarrow 243 = \alpha^5 + \beta^5 + 25(-18) + 10(25)(3)$$

$$\Rightarrow \alpha^5 + \beta^5 = -57$$

$$\frac{\beta\alpha^6 + \alpha\beta^6}{\alpha\beta} = \frac{\alpha\beta(\alpha^5 + \beta^5)}{\alpha + \beta} = \frac{5(-57)}{3} = -95$$

4. **Ans. 8.00**

Case-I : $T_1 S S T_2 S S T_3 S S T_4 S S$

Case-II : $S S T_1 S S T_2 S S T_3 S S T_4$

Case-III : $S T_1 S S T_2 S S T_3 S S T_4 S$

$$\text{Total ways} = 3 \times 4! \times 8! = 72 \times 8! = 8(9!)$$

5. **Ans. 7.00**

$$x + y = 14$$

$$\Delta^2 = 9 \times 5 \times (9 - x)(9 - y)$$

$$\sqrt{(9-x)(9-y)} \leq \frac{18-(x+y)}{2}$$

$$\sqrt{(9-x)(9-y)} \leq 2$$

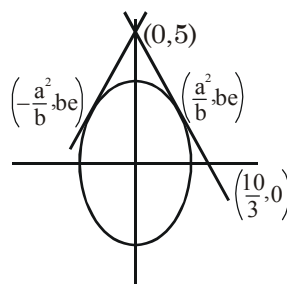
$\Rightarrow$ for maximum area $x = y$.

$$\therefore x = y = 7$$

$$\Rightarrow \frac{x+y}{2} = 7$$

6. **Ans. 3.70**

Sol. Let ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$



$$\text{tangent } \frac{x}{b} + \frac{ye}{b} = 1$$

$$\therefore \frac{b}{e} = 5$$

$$\text{and } b = \frac{10}{3} \Rightarrow e = \frac{2}{3}$$

$$\Rightarrow a^2 = \frac{500}{81} \therefore \text{L.R} = \frac{2a^2}{b} = \frac{100}{27}$$

7. **Ans. 1.41 or 1.42**

$$\text{Tangent to } y^2 = 8x \text{ is } y = mx + \frac{2}{m} \quad \dots(i)$$

$$\text{Tangent to } \frac{x^2}{1} - \frac{y^2}{3} = 1$$

$$\text{is } y = mx + \sqrt{m^2 - 3} \quad \dots(ii)$$

compare (i) and (ii), we get

$$c = \sqrt{2}$$

8. **Ans. 1.00**

$$\text{Let } \alpha + \beta = 2\gamma \quad \dots(i)$$

$$\text{where } \tan \alpha = \frac{2x+1}{x+1}, \tan \beta = \frac{2x-1}{x-1} \text{ \&}$$

$$\tan \gamma = x + 1$$

Taking tan in (i) we get

$$\frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta} = \frac{2 \tan \gamma}{1 - \tan^2 \gamma}$$

$$\Rightarrow \frac{\frac{2x+1}{x+1} + \frac{2x-1}{x-1}}{1 - \frac{4x^2-1}{x^2-1}} = \frac{2(x+1)}{1 - (x+1)^2}$$

$$\Rightarrow \frac{2x^2 - x - 1 + 2x^2 + x - 1}{x^2 - 1 - 4x^2 + 1} = \frac{2(x+1)}{-x^2 - 2x}$$

$$\Rightarrow \frac{2(2x^2 - 1)}{-3x^2} = \frac{2(x+1)}{-(x^2 + 2x)}$$

$$x = 0$$

$$\text{or } 2x^3 + 4x^2 - x - 2 = 3x^2 + 3x$$

$$\Rightarrow 2x^3 + x^2 - 4x - 2 = 0$$

$$\Rightarrow x^2(2x + 1) - 2(2x + 1) = 0$$

$$\Rightarrow (x^2 - 2)(2x + 1) = 0$$

$$\Rightarrow x = \sqrt{2}, -\sqrt{2} \text{ or } x = -\frac{1}{2}$$

$x = -\sqrt{2}$ is rejected $\because$ it does not satisfy (i)

$$\Rightarrow x_1 = -\frac{1}{2}, x_2 = 0 \text{ \& } x_3 = \sqrt{2}$$

$$\Rightarrow 2x_1 + x_2 + x_3^2 = 1$$



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2020 - 2021)

JEE(Advanced)

REVIEW TEST

08-11-2020

JEE(Main + Advanced) : NURTURE COURSE (PHASE : TEPS)

PAPER-2

PART-2 : CHEMISTRY

SOLUTION

SECTION-I

1. Ans. (A,C)
2. Ans. (A,C)
3. Ans. (A,B,C,D)
4. Ans. (A)

$$P_d = P_A^0 Y_A + P_B^0 Y_B$$

$$P_d = 100 \times \frac{4}{22} + 300 \times \frac{18}{22}$$

$$= 263.64$$

$$Y_A = \frac{100 \times 0.4}{100 \times 0.4 + 300 \times 0.6} = \frac{4}{22}$$

5. Ans. (A,D)
6. Ans. (B,C)

7. Ans. (D)
8. Ans. (A)
9. Ans. (A)
10. Ans. (C)

SECTION-II

1. Ans. (15.00)
2. Ans. (5.00)
3. Ans. (10.00)
4. Ans. (17.50)
5. Ans. (5.00)
6. Ans. (3.00)
7. Ans. (6.00)
8. Ans. (2.00)

PART-3 : MATHEMATICS

SOLUTION

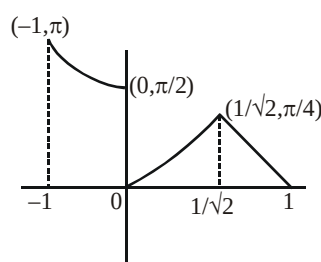
SECTION-I

1. Ans. (A,B,C)

$$\text{Sol. } \frac{2 \tan \frac{x}{2}}{1 + \tan^2 \frac{x}{2}} + \frac{\left(1 - \tan^2 \frac{x}{2}\right)}{\left(1 + \tan^2 \frac{x}{2}\right)} = -\frac{1}{3}$$

$$\text{Let } \tan \frac{x}{2} = t \text{ and solve}$$

2. Ans. (A,D)



$$\lambda_1 = 2, \lambda_2 = 2, \lambda_3 = 1$$

3. Ans. (A,B,D)

$$\text{Sol. } S_n = \sum_{r=2}^n \left[\frac{3^r}{r} - \frac{3^{r-1}}{(r-1)} \right] = \frac{3^n}{n} - 3$$

$$(A) (B) S_9 = 3^7 - 3 = 3(27^2 - 1) = 3.26.28$$

$$(C) S_{10} = \frac{3^{10}}{10} - 3 \Rightarrow 10S_{10} + 30 = 3^{10}$$

$$(D) n(S_n + 3) = 3^n$$

so $3^7, 3^{10}, 3^{13}$ are in G.P.

4. Ans. (A,B,C)

Obviously all options are correct except (D)

5. Ans. (A,C,D)

$$(1+x)^{100} = C_0 + C_1x + C_2x^2 + C_3x^3 + \dots + C_{100}x^{100}$$

and differentiation $100(1+x)^{99}$

$$= C_1 + 2C_2x + 3C_3x^2 + \dots + 100C_{100}x^{99} \dots (i)$$

$$(x+1)^{100} = C_0x^{100} + C_1x^{99} + C_2x^{98} + \dots + C_{100}x^0 \dots (ii)$$

in(i) $\times$ (ii) coeff of x^{99} is

$$100 {}^{199}C_{100} = C_1^2 x^{99} + 2C_2^2 x^{99} + \dots + 100C_{100}^2$$

$$100 {}^{199}C_{100} = 100 \cdot {}^{199}C_{99}$$

$$= 100 \times \frac{199!}{99!100!} = \frac{100}{200} \times \frac{200!}{100!100!} \times 100 = 50 {}^{200}C_{100}$$

$$50 {}^{200}C_{100} = 50 \cdot \frac{1.2.3.4 \dots 200}{100!100!}$$

$$= 50 \frac{[1.3.5 \dots 199]}{100!} 2^{100} \frac{(1.2.3 \dots 100)}{100!}$$

6. Ans. (A,B,C,D)

Sol. Given minor axis $\Rightarrow 2y + x = 3$

Equation of minor axis $= 2x - y = k$

Passes through (2, 3) $k = 1$

hence major axis $2x - y = 1$

centre of ellipse (1, 1)

$$\frac{a}{e} = \sqrt{(3-1)^2 + (2-1)^2} = \sqrt{5}$$

$$\therefore b^2 = a^2(1 - e^2)$$

$$\therefore \frac{5}{4} = 5e^2(1 - e^2) \Rightarrow e = \frac{1}{\sqrt{2}}$$

$$\text{L.R.} = \frac{\sqrt{5}}{\sqrt{2}}$$

equation of ellipse

$$4(2x - y - 1)^2 + 2(x + 2y - 3)^2 = 25$$

7. Ans. (A)

Sol. (A) $\cos A = \frac{4 + 3 - a^2}{2 \cdot 2 \cdot \sqrt{3}} = \frac{\sqrt{3}}{2}$

$$7 - a^2 = 6$$

$$a^2 = 1$$

$$a = 1$$

$$\frac{a}{\sin A} = 2R \Rightarrow 2R = 2 \Rightarrow R = 1$$

(B) $\Delta = \frac{9\sqrt{3}}{2}$

$$\Rightarrow bc = 18 \Rightarrow b - c = 3\sqrt{3} \Rightarrow b^2 + c^2 = 63$$

$$\cos A = -\frac{1}{2}$$

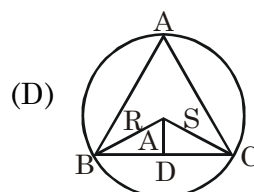
$$\frac{b^2 + c^2 - a^2}{2bc} = -\frac{1}{2}$$

$$a = 9$$

(C) $\frac{\Delta}{s-a} = r_1 \Rightarrow s-a=10$

$$\frac{\Delta}{s-b} = 50 \Rightarrow s-b=2$$

$$b-a=8$$



$$r = SD = R \cos A$$

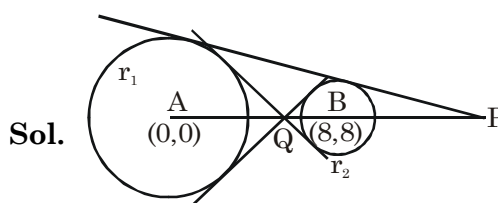
$$\cos A = \frac{r}{R} = 4 \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2}$$

$$1 - 2 \sin^2 \frac{A}{2} = 4 \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2}$$

$$1 = 2 \sin \frac{A}{2} \cos \frac{B-C}{2}$$

$$2 \cos \frac{B+C}{2} \cos \frac{B-C}{2} = 1 \Rightarrow \cos B + \cos C = 1$$

8 Ans. (B)



Sol.

Let A and B be centres of circles and P and Q be intersection of direct and transverse common tangent

Now $\frac{AQ}{BQ} = \frac{r_1}{r_2} = \frac{2}{1}$ (internally)

$$\therefore Q \left(\frac{2(8) + 1(0)}{2+1}, \frac{2(8) + 1(0)}{2+1} \right) = \left(\frac{16}{3}, \frac{16}{3} \right)$$

$$\frac{AP}{BP} = \frac{2}{1} \text{ (Externally)}$$

$$\therefore P \left(\frac{2(8) - 1(0)}{2-1}, \frac{2(8) - 1(0)}{2-1} \right) = P(16, 16)$$

Let d = distance between centres $= \sqrt{128}$

length of direct common tangent

$$= \sqrt{d^2 - (r_1 - r_2)^2} = \sqrt{128 - 1} = \sqrt{127}$$

Length of transverse tangent

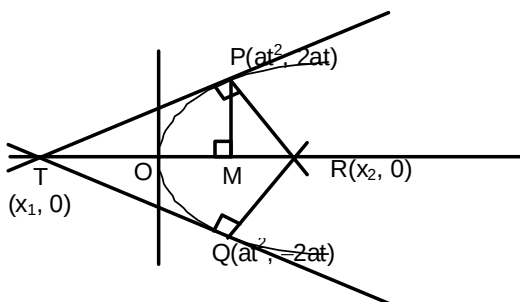
$$= \sqrt{d^2 - (r_1 + r_2)^2}$$

$$= \sqrt{128 - 9} = \sqrt{119}$$

9. Ans. (B)

(A). $y^2 = 4x$... (1)

Clearly, if P (at^2 , $2at$) then by symmetry,



Q (at^2 , $-2at$)

Equation of tangent is

$$ty = x + at^2$$

for T, $y = 0$, $x_1 = -at^2$

and equation of normal is

$$y = -tx + 2at + at^3$$

for R, $y = 0$, $x_2 = (2a + at^2)$... (2)

Here, $a = 1 \Rightarrow x_1 = -t^2$ and $x_2 = (2 + t^2)$

$$x_2 = 3 = 2 + t^2 \Rightarrow t^2 = 1 \Rightarrow t = \pm 1$$

Take $t = 1$, then $x_1 = -1$,

$$\therefore PM = 2at = 2 \times 1 \times 1 = 2$$

$$RT = |x_1 - x_2| = (1 + 3) = 4$$

$\therefore$ Area of quadrilateral PTQR

$$= 2 \times \left(\frac{1}{2} \times 4 \times 2 \right) = 8 \text{ sq. units.}$$

(C) Clearly, RT will be the diameter of circle
 $\therefore$ Circumference = ($\pi \times$ diameter)

$$\frac{\text{Circumference}}{4\pi} = \frac{\pi \times RT}{4\pi} = \frac{\pi \times 4}{4\pi} = 1.$$

(D) Since in the part (1),
 we have found $x_2 (2a + at^2) > 2a$, (if $t \neq 0$)
 $\therefore$ For three real normals, $x_2 > 2a = 2 \times 1$
 $= 2$
 i.e. $x_2 > 2$.

10. Ans. (B)

(A) $f(x) = x \sin x$

$$f(-x) = f(x) \Rightarrow f \text{ is even}$$

$$\lim_{x \rightarrow \infty} f(x) \rightarrow \infty$$

$\Rightarrow$ Unbounded & aperiodic.

(B) $f(x) = x + \tan^{-1}x$

$$f(-x) = -f(x) \Rightarrow 'f' \text{ is odd}$$

$$\lim_{x \rightarrow \infty} f(x) \rightarrow \infty$$

$\Rightarrow$ 'f' is unbounded and aperiodic

(C) Bounded, periodic and even

(D) Bounded periodic, neither odd nor even.

SECTION-II

1. Ans. 0.00

$$2^{4x^2-1} - 2^{x^4} > 0$$

$$2^{4x^2-1} > 2^{x^4}$$

$$4x^2 - 1 > x^4$$

$$x^4 - 4x^2 + 1 < 0$$

$$2 - \sqrt{3} < x^2 < 2 + \sqrt{3}$$

(i) $x^2 > 2 - \sqrt{3}$

$$x^2 - \left(\frac{\sqrt{3}-1}{\sqrt{2}} \right)^2 > 0$$

$$x \in \left(-\infty, -\frac{(\sqrt{3}-1)}{\sqrt{2}} \right) \cup \left(\frac{\sqrt{3}-1}{\sqrt{2}}, \infty \right) \dots (i)$$

(2) $x^2 < 2 + \sqrt{3}$

$$x^2 - \left(\frac{\sqrt{3}+1}{\sqrt{2}} \right)^2 < 0$$

$$x \in \left(-\frac{(\sqrt{3}+1)}{\sqrt{2}}, \frac{\sqrt{3}+1}{\sqrt{2}} \right) \dots (ii)$$

$$\Rightarrow x \in \left(-\frac{\sqrt{3}+1}{\sqrt{2}}, -\frac{(\sqrt{3}-1)}{\sqrt{2}} \right) \cup \left(\frac{\sqrt{3}-1}{\sqrt{2}}, \frac{\sqrt{3}+1}{\sqrt{2}} \right)$$

$$\Rightarrow a + b + c + d = 0$$

ALLEN**2. Ans. 2.00**

$$f(x) = x^2 \operatorname{sgn} x$$

$$\text{Now, } f(x+a) \geq 2f(x) \quad \forall x \in [a, a+2]$$

$$\text{Case: I If } a+2 < 0 \Rightarrow a < -2$$

$$\Rightarrow (x+a)^2 \operatorname{sgn}(x+a) \geq 2x^2 \operatorname{sgn} x$$

$$\text{Now, } \operatorname{sgn} x < 0 \text{ \& } \operatorname{sgn}(x+a) < 0$$

$$(x+a)^2 \leq 2x^2 \Rightarrow x^2 - 2ax - a^2 \geq 0$$

$$\text{which is not possible } \forall x \in [a, a+2]$$

$$\text{Case : II } a < 0 < a+2$$

$$\text{Put } x = 0 \quad f(a) \geq 2f(0)$$

$$-a^2 \geq 0 \text{ Not possible}$$

$$\text{Case : III } a > 0 \therefore (x+a)^2 \operatorname{sgn}(x+a) \geq 2x^2 \operatorname{sgn} x$$

$$\Rightarrow x^2 - 2ax - a^2 \leq 0$$

$$\therefore f(a) \leq 0$$

$$\Rightarrow -2a^2 \leq 0 \Rightarrow \text{true}$$

$$f(a+2) \leq 0$$

$$a^2 + 4a + 4 - 2a^2 - 4a - a^2 \leq 0$$

$$a^2 \geq 2 \Rightarrow a \in [\sqrt{2}, \infty]$$

3. Ans. 125.00

$$\text{Sol. } C_1 \rightarrow C_1 - \sqrt{\beta}C_3, C_2 \rightarrow C_2 + \sqrt{\alpha}C_3$$

$$\Delta = \begin{vmatrix} 1+\alpha+\beta & 0 & -2\sqrt{\beta} \\ 0 & 1+\alpha+\beta & 2\sqrt{\alpha} \\ (1+\alpha+\beta)\sqrt{\beta} & -\sqrt{\alpha}(1+\alpha+\beta) & 1-\alpha-\beta \end{vmatrix}$$

$$= (1+\alpha+\beta)^2 \begin{vmatrix} 1 & 0 & -2\sqrt{\beta} \\ 0 & 1 & 2\sqrt{\alpha} \\ \sqrt{\beta} & -\sqrt{\alpha} & 1-\alpha-\beta \end{vmatrix}$$

$$= (1+\alpha+\beta)^2 [1+\alpha-\beta+2\beta]$$

$$= (1+\alpha+\beta)^3 = 125$$

4. Ans. 2.00

$$\sin(\cot^{-1}(\cos C \tan^{-1} x))$$

$$= \sin \left(\cot^{-1} \cos \cos^{-1} \left(\frac{1}{\sqrt{1+x^2}} \right) \right)$$

$$= \sin \left(\cot^{-1} \left(\frac{1}{\sqrt{1+x^2}} \right) \right) = \sin^{-1} \left(\tan^{-1} \sqrt{1+x^2} \right)$$

$$= \sin^{-1} \left(\sin^{-1} \left(\frac{\sqrt{1+x^2}}{\sqrt{2+x^2}} \right) \right) = \sqrt{\frac{1+x^2}{2+x^2}}$$

$$\Rightarrow \sqrt{\frac{1+x^2}{2+x^2}} = a \Rightarrow x^2 = \frac{2a^2-1}{1-a^2}$$

$$\Rightarrow a \neq \pm 1$$

$$(1-a^2)(2a^2-1) \geq 0$$

$$\Rightarrow \frac{1}{2} \leq a^2 < 1 \Rightarrow a \in \left[\frac{1}{\sqrt{2}}, 1 \right)$$

$$p = \frac{1}{\sqrt{2}}, q = 1 \Rightarrow q + \frac{1}{p} = 1 + \sqrt{2}$$

$$\Rightarrow \left[q + \frac{1}{p} \right] = 2$$

5. Ans. 1.00

$$\tan^{-1} \left(\frac{1}{3} \right) + \tan^{-1} \left(\frac{2}{9} \right) + \tan^{-1} \left(\frac{4}{33} \right) + \dots$$

$$t_n = \tan^{-1} \left(\frac{2^{n-1}}{1+2^{2n-1}} \right) = \tan^{-1} \left(\frac{2^n - 2^{n-1}}{1+2^n \cdot 2^{n-1}} \right)$$

$$= \tan^{-1}(2^n) - \tan^{-1}(2^{n-1})$$

6. Ans. 8.00

$$(1-y)^m(1+y)^n$$

$$= \left(1 - my + \frac{m(m-1)}{2}y^2 + \dots \right) \left(1 + ny + \frac{n(n-1)}{2}y^2 + \dots \right)$$

$$= 1 + (n-m)y + \left(\frac{m(m-1)}{2} + \frac{n(n-1)}{2} - mn \right) y^2 + \dots$$

$$\Rightarrow n - m = 10$$

$$\& \frac{m(m-1)}{2} + \frac{n(n-1)}{2} - mn = 10$$

$$\Rightarrow m + n = 80$$

7. **Ans. -1.00**

$$(1 + x + x^2 + \dots + x^9)^{-1}$$

$$= \left(\frac{x^{10} - 1}{x - 1} \right)^{-1} = (1 - x)(1 - x^{10})^{-1}$$

$$= (1 - x) \sum_{r=0}^{\infty} x^{10r}$$

$$\text{Coefficient } x^{401} = -1$$

8. **Ans. 24.00**

$$x_{\max.} = 999$$

$$\therefore x + 32 = 1031$$

$$\text{Now } (x + 32)_{\min} = 1000$$

$$\Rightarrow 1000 < x + 32 < 1031$$

$$\Rightarrow x + 32 = 1001 \Rightarrow x = 969$$

$$\Rightarrow \text{sum of digits} = 24$$